

Supervisor briefing — update since the last meeting

Can a soundtrack's emotion predict a film's genre? This page covers only what happened since the fourth meeting (24 September 2026): the six notes from that meeting, what came of them, and the work on the thesis text that followed (state: 7 October 2026). Earlier rounds, the three routes (direct audio, emotion route, PCA-8 control), the definitions, every result table and the previous Q&A are in `briefing_full.md`; every experiment in detail is in the technical log, `docs/README.md` (section 22 for this round).

1. Your notes from the last meeting

#	Your note	What was done	Result
1	Explain the F1 metric more; is there a better approach?	Every route scored under 13 metrics on the same predictions (section 2)	Macro-F1 stays the right headline , but the in-domain lead depends on the decision threshold
2	Standard deviation / variance of the cross-validation: stable and reproducible?	Spread over repeats, folds, films and five master seeds (section 3)	Stable and reproducible ; the emotion lead is positive under every seed
3	Box office vs the Blockbuster dataset, under all metrics	Worldwide gross for all 110 films; 6 per-film metrics, emotions, genre, prediction, all 140 MIR features (section 4)	Classification quality: no relation . Emotions: only through the budget . Then left out of the thesis (7 Oct)
4	Write the rest of the chapters	Full drafts of every remaining chapter and the abstract (section 5)	On Overleaf, compiles with 0 errors ; reviewed, style-revised, references checked. Abstract, conclusions and contributions held back for the author
5	Results as graphs, not only tables	Figures drawn directly from the results files (section 6)	Eight figures in the Evaluation chapter, with short captions
6	Figures from other papers are allowed, with the source	Noted (section 6)	The circumplex figure could now be the original

2. Note 1 — the F1 metric, and whether there is a better one

What F1 measures

For **one genre**, say Horror:

- **Precision** = of the clips the model *called* Horror, the share that really are.
- **Recall** = of the clips that really *are* Horror, the share the model found.
- **F1** = the harmonic mean of the two, $2 \cdot P \cdot R / (P + R)$. It is high only when both are high: a model that calls everything Horror has recall 1 but poor precision, and one that calls almost nothing Horror has the opposite problem.

A clip can have several genres, so there is one F1 per genre, and they have to be combined into one number. That is where the variants differ:

averaging	how	what it rewards
macro-F1 (ours)	F1 per genre, then the plain mean	every genre counts the same: Horror (28 clips) as much as Drama (239)
micro-F1	pool all clip-genre decisions, one F1	dominated by the frequent genres

weighted-F1	F1 per genre, mean weighted by genre size	in between
samples-F1	F1 per clip over its genre list, then the mean	dominated by the frequent genres

What the alternatives show

Same predictions, 5 genres, mean of 10 repeats (`results/metrics_stability.json`). "Random" is a base-rate random guess, "most freq." always predicts the most common genres (in practice: Drama).

metric	emotion route	direct VGGish	direct AST	PCA-8 control	most freq.	random
macro-F1 (headline)	0.417	0.377	0.371	0.372	0.168	0.308
macro-F1, threshold tuned per genre	0.440	0.429	0.427	0.441	0.168	0.308
macro average precision (no threshold)	0.372	0.361	0.359	0.356	0.309	0.321
macro ROC-AUC (no threshold)	0.608	0.567	0.571	0.567	0.500	0.501
macro-F1 per film (clips averaged)	0.469	0.375	0.307	0.418	0.169	0.301
micro-F1	0.482	0.471	0.515	0.446	0.571	0.477
exact match	0.082	0.105	0.170	0.095	0.283	0.133
Hamming loss (lower is better)	0.404	0.359	0.314	0.393	0.218	0.323

Three findings.

1. **Macro-F1 is the right headline.** Micro-F1, samples-F1, exact match, Hamming loss and Jaccard are all *won by the most-frequent baseline*, a model that has learned nothing but "say Drama". A metric that ranks it first cannot answer a question about Horror or Comedy.
2. **Among the metrics that treat genres equally, the emotion route is first:** macro-F1, weighted-F1, both threshold-free metrics and the film-level score (0.469 vs 0.375, $p = 0.041$).
3. **But the in-domain lead depends on where the probability is cut.** Our classifiers say "yes, this genre" when their probability exceeds 0.5. If each genre's cut is instead tuned on the training data, every route improves and they converge (0.427–0.441). The emotion lead is then no longer significant (+0.011 vs VGGish, $p = 0.44$; level with PCA-8). The threshold-free metrics, which measure only how well clips are *ranked*, still put the emotion route first, but not significantly. At the default cut the emotion route wins mainly through **recall** (0.596 vs 0.439).

The honest statement for the thesis: in-domain, the emotion route loses nothing against direct audio and is ahead at the standard operating point; it is not better under every way of scoring. The **zero-shot** result (0.511 vs 0.407) is a different situation: there are no labels on the target dataset to tune a threshold with. Whether that advantage survives a threshold tuned on the *source* dataset has not been tested yet; it is the one open check this raises (section 7).

3. Note 2 — standard deviation, stability, reproducibility

Macro-F1, 5 genres. Three sources of variation, measured separately:

route	SD over the 10 repeats (fold assignment)	SD over the 50 single folds	film bootstrap SE (which films)	SD of the mean over 5 seeds
human ratings (ceiling)	0.009	0.057	0.027	0.002
emotion route	0.011	0.054	0.024	0.003
direct VGGish	0.020	0.057	0.023	0.008

direct AST	0.023	0.050	0.018	0.001
PCA-8 control	0.015	0.061	0.026	0.004

- **Repeats** vary little (≤ 0.023); the emotion route is among the steadiest.
- **Single folds** vary a lot (≈ 0.055), because a fold holds only about 8 films. That is why one 5-fold run cannot resolve differences of 0.04, and why we repeat 10 times.
- **Which films are in the dataset** is the largest source (0.018–0.027). Only more films can reduce it.
- **Seeds.** The whole 10×5 protocol was re-run under five master seeds (42, 1, 2, 3, 4), which changes both the fold assignment and the random forest. The emotion route scores 0.410–0.417 under every seed. Its lead over direct VGGish is positive under **every** seed (+0.037 to +0.059), as are its leads over AST and PCA-8. The PCA-8-vs-VGGish difference flips sign between seeds (−0.005 to +0.012), which is what "no effect" looks like.
- **Reproducibility.** Under seed 42 the new script reproduces the per-repeat scores of the main analysis (cv_corrected.json) exactly, for every route.

Figure: figures/stability_repeats.pdf.

4. Note 3 — box office on Blockbuster (done, then left out of the thesis)

Data. Blockbuster ships only title slugs, so each film was matched to Wikidata by title and release year, and its **worldwide** gross taken from Box Office Mojo: all 110 films, one source, USD 9 million to 2.8 billion. The budget was found for 84 films. Blockbuster has no ratings, so the emotions are those the Eerola-trained model predicts per cue. Every correlation is Spearman with a permutation p-value and a Benjamini–Hochberg correction for multiple tests.

question	answer
Is a film's genre easier to predict when it grossed more? 6 per-film metrics × every route, in-domain and zero-shot: 42 tests	No. Nothing survives the correction; the largest correlation is $p = +0.24$
Do the soundtrack's emotions track gross?	Yes, at first sight: anger $p = +0.43$, tension $+0.36$, fear $+0.30$, tenderness -0.37 , valence -0.32 , all significant after correction. With the budget controlled, every one is within ± 0.11.
Does genre?	Action (median \$475M vs \$129M) and Sci-Fi (\$615M vs \$159M) gross more; Drama and Comedy less
Can gross be predicted from the soundtrack?	Partly (emotion $R^2 = 0.28$, VGGish 0.35), but the budget alone does better (0.49) , and adding the emotions to it does not help (0.45)

Reading: blockbuster music mirrors the *scale and kind of production*. Expensive action and sci-fi films are scored with tense, angry, energetic music and earn more; budget and gross themselves correlate at $p = +0.78$. Once the budget is known, the soundtrack adds nothing. Exploratory, not causal.

A bug found on the way. The Box Office Mojo parser read the navigation menu and stored US-only figures. Fixed; on Eerola 7 grosses changed and every conclusion stayed the same (quality vs gross $p = +0.197$, $p = 0.246$).

Decision (7 Oct): the box-office analysis is **not in the thesis**: it answers a question about commercial success, not about genre. The text is kept in docs/latex/held_back/box_office.tex and the analysis in the repository.

5. Note 4 — the chapters (now on Overleaf)

All chapters are written and **on Overleaf**; the thesis compiles with **0 errors**.

chapter	state
1 Introduction	drafted; the list of contributions is held back for the author to write
2 Fundamentals, 3 Related Work	written earlier; revised
4 Methods	drafted (datasets, three label spaces, six representations, Stage 1 with the three derived features, Stage 2, baselines, protocol, implementation; TikZ pipeline figure)
5 Experiments and Evaluation	drafted, with the result figures and the new section on metric and

	stability
6 Discussion	drafted
7 Conclusions and Future Work, abstract	held back: only headings and placeholders on Overleaf; drafts in docs/latex/held_back/

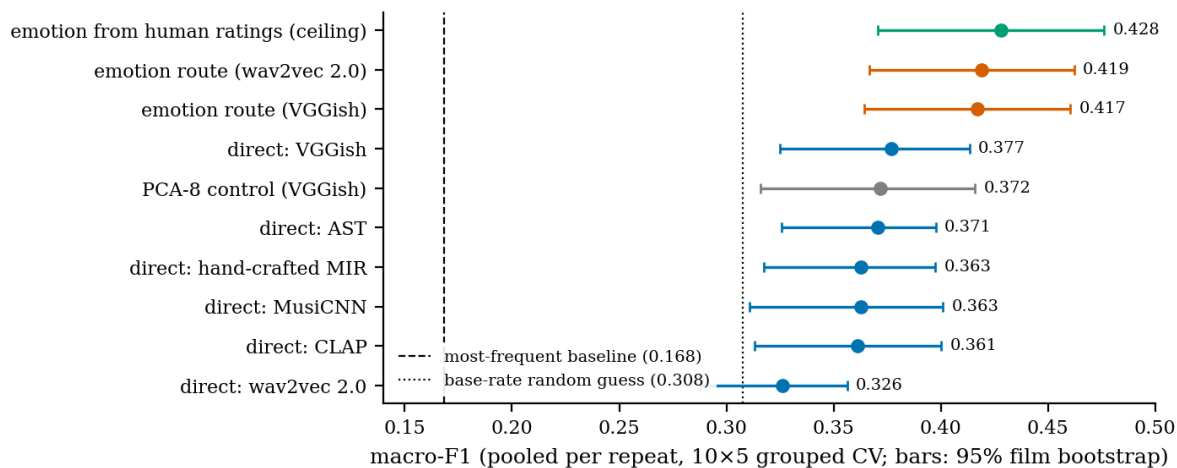
What was done to the text after drafting:

1. **Missing references added** (librosa, scikit-learn, PyTorch, transformers, Lipton et al. 2014 on F1 thresholds; LAION-CLAP, see below).
2. **An examiner-style review** found factual slips, all corrected:
 - CLAP was cited to the wrong model (the checkpoint used is LAION-CLAP).
 - VGGish was said to be trained on AudioSet (it was trained on YouTube videos and is distributed with AudioSet).
 - The reliability ceiling 0.897 is an intraclass correlation, not r^2 .
 - Clip length, film counts and dataset size were stated wrongly in places.
 - One argument was circular (the scaling ablation), one compared margins across label spaces that are not comparable.
 - A limitation was added: the five-genre subset was chosen partly by emotional signature.
3. **A consistency pass** over the whole thesis: about 25 wording and pointer fixes (numbers stated two ways, wrong cross-references, British spelling).
4. **Style revision** of the prose of every chapter (plainer sentences, fewer set phrases and dashes). A script checked that no number, reference, citation, table or figure changed.
5. **Captions shortened:** every figure and table has a one-line caption; the legend (colours, whiskers, columns) moved into the text after the first mention.
6. **Every reference and link checked:** all 36 bibliography entries against Crossref and OpenAlex (all real, all matching); the GitHub link and the three Hugging Face models resolve.
7. **AI-usage declaration filled in** from the repository and Overleaf history: models Claude Opus 5 and Claude Opus 5.5, July to October 2026, which parts were drafted, edited or generated. Three red notes remain that only the author can fill.

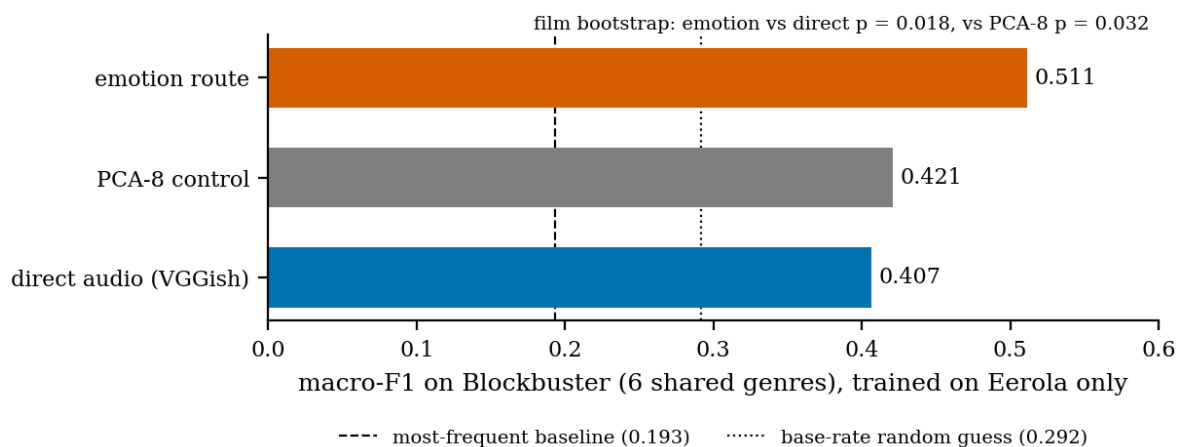
6. Notes 5–6 — figures

Eight figures, drawn by `experiments/make_figures.py` directly from the results files, so a figure can never disagree with a table.

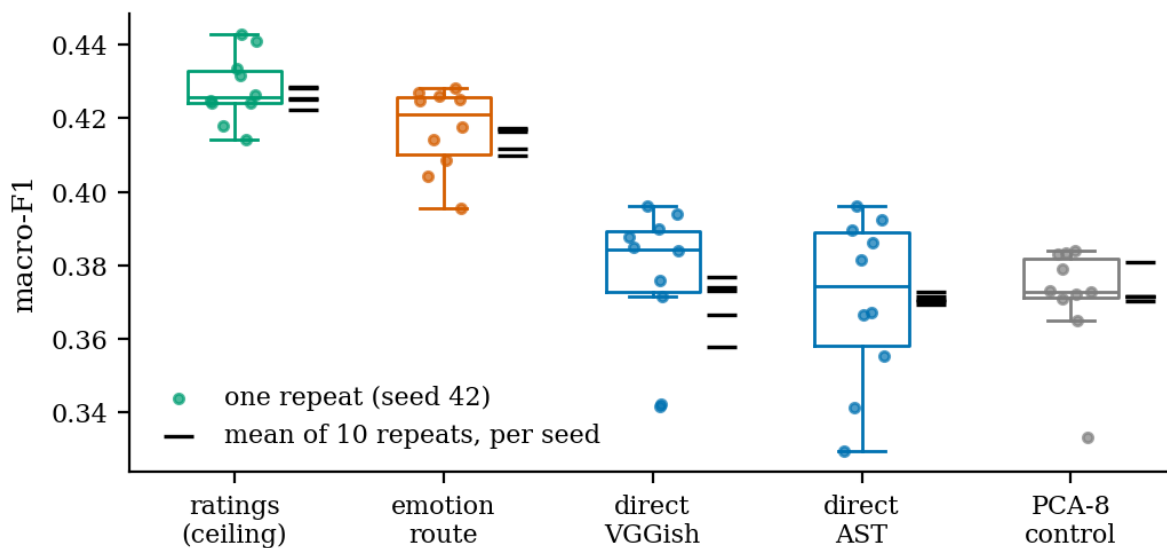
figure	shows
indomain_eerola5	every representation and route on 5 genres, with 95 % intervals and both chance levels
stability_repeats	the 10 repeats per route, and the mean under each of 5 seeds
metric_family	the routes under 7 metrics, as distance above random guessing
stage1_r2	emotion-prediction R^2 per emotion $\times$ representation (heatmap)
ablation	removing each emotion: difference to all eight, with intervals
signatures	each genre's emotional signature, Eerola vs Blockbuster side by side
zero_shot	Eerola $\rightarrow$ Blockbuster: emotion 0.511 vs PCA-8 0.421 vs direct 0.407
cross_dataset	both transfer directions and the pooled design



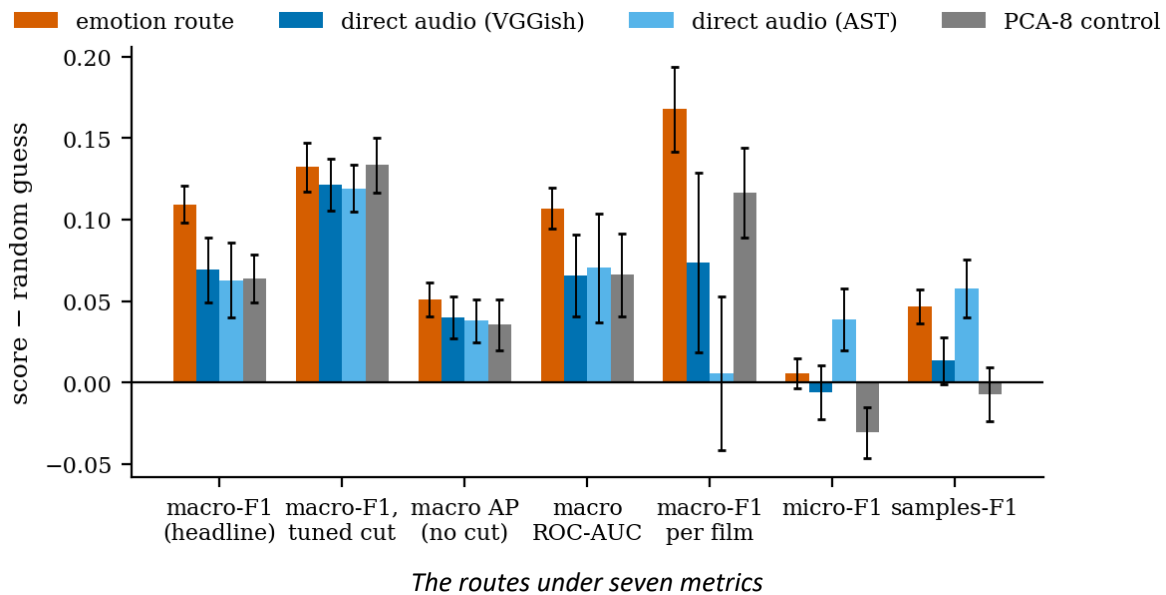
In-domain, five genres: every route with 95 % intervals



Zero-shot transfer from Eerola to Blockbuster



Stability over repeats and seeds



Figures from other papers. The circumplex figure in Fundamentals is redrawn "after Russell (1980)"; the original could replace it with its source. Ma et al. (2021) is published under CC BY 4.0, so its figures may be reused with attribution.

7. A correction since: the pooled cross-dataset design

The results file of the pooled design (both datasets in training) came from a 3×5 run, while the script and the text said 5×5. It was re-run at 5×5 (7 Oct). The two transfer directions reproduce exactly; the pooled numbers move slightly and the conclusion stays:

pooled design	before (3×5)	now (5×5)
emotion route / direct / PCA-8	0.424 / 0.421 / 0.398	0.432 / 0.428 / 0.405
emotion vs direct	+0.003, p = 0.906	+0.004, p = 0.850

Once both datasets are in training, the two routes tie: the emotion advantage is a generalisation advantage.

8. Open items

- 1. Write the held-back parts:** abstract, Chapter 7 (Conclusions and Future Work) and the list of contributions. Future Work must keep the experiment the Discussion points to (below).
- 2. Zero-shot with a threshold tuned on the source dataset:** the one check the metric analysis raises that has not been run.
- 3. AI-usage declaration:** three red notes (other tools, how the drafts were revised, the level for the held-back parts) and the date.
- 4. Read the whole thesis once,** using docs/verification_checklist.md: what to check, the numbers that no results file stores, every correction made, and what to look at in the PDF (the two TikZ diagrams have not been checked visually).

9. Questions to expect, and the answers

Why macro-F1 and not accuracy or Hamming loss? Because on this data those are won by a model that always says "Drama": exact match 0.283 and Hamming loss 0.218, better than every real model. Macro-F1 gives each genre the same weight, so it cannot be won by ignoring the rare ones.

Is the emotion route better in-domain or not? At the standard decision threshold, yes: ahead of every direct representation in all 10 repeats and all 5 seeds, significant under the film bootstrap ($p = 0.042$) and borderline under Nadeau–Bengio ($p = 0.151$). With per-genre thresholds tuned on the training data, the routes converge and the lead is not significant. So in-domain it "loses nothing and is ahead at the default operating point", not "better under every metric".

Then what is left of the main claim? The cross-dataset result: 0.511 vs 0.407 ($p = 0.018$) from Eerola to Blockbuster, the reverse direction agreeing ($p < 0.001$), a tie once both datasets are in training ($p = 0.850$), and interpretable predictions whose emotional signatures replicate on a second dataset ($r = 0.836$). The advantage is a generalisation advantage; the thesis says so.

Would threshold tuning also erase the zero-shot advantage? Unknown: not tested yet (open item 2).

How stable are the numbers? Very: the 10-repeat mean moves by at most 0.008 between seeds, and the emotion lead over VGGish stays between +0.037 and +0.059 under every seed. The main uncertainty is which 41 films the dataset contains (bootstrap SE about 0.02).

Why is the box-office analysis not in the thesis? It was done (section 4) and its answer is clear: classification quality does not relate to gross, and the emotions of the score relate to it only through the budget. It answers a question about commercial success rather than about genre, so it was left out.

How was AI used? As a programming and writing assistant (Claude Opus 5 and 5.5): code, documentation, chapter drafts, figures and reference checking. The declaration at the end of the thesis lists every part and its level; the research questions, datasets and decisions are the author's.

Headline numbers as they stand

Each row is its own label space, compared only against its own chance level (base-rate random guess):

setting	emotion route	best direct audio	random guess
Eerola, 5 genres, in-domain (default threshold)	0.417	0.377	0.308
Eerola, 5 genres, in-domain (tuned thresholds)	0.440	0.429	0.308
Eerola → Blockbuster, zero-shot (6 genres)	0.511	0.407	0.292
Blockbuster, in-domain (6 genres)	0.616	0.593	0.295

In-domain significance (default threshold): emotion vs VGGish film bootstrap $p = 0.042$, Nadeau–Bengio $p = 0.151$. Zero-shot: +0.106 [+0.015, +0.186], $p = 0.018$. On 8 genres the pipeline scores 0.318 against 0.276 for AST (most-frequent baseline 0.102). The 5-genre random guess is 0.308, the simulated value stored in `metrics_stability.json`.